

Introduction to .NET Development Environment

Objectives:

- Integrated Development Environment
- Components of IDE
- Templates Supported
- Writing a simple application in VB.NET

2.1 VISUAL STUDIO.NET INTEGRATED DEVELOPMENT

This IDE is provided with a common interface for developing various types of applications. It acts as a centralized repository for the application, writing, compiling and debugging the code. Each and every visual studio uses the different versions of framework. The main versions are VS2002, VS2003, VS2005 and VS2008.

Visual studio.Net and framework differs from each other in some aspects. Visual Studio is integrated with editor and compiler facility also. So users can easily write and compile the code using the visual studio. Otherwise we can write the code in the notepad or any other editors like Java. Developers can compile these code using commands in command prompt. It would be very difficult to remember the commands. Visual Studio is enriched with all GUI features, drag and drop facility, navigational facility, compilation and execution environment also. The IDE provides the rich environment for writing, debugging, compiling and executing the application.

But for running the application framework has to be installed in the machine. The application cannot be executed without framework. But application can execute without Visual Studio. Installation of the Visual Studio includes the framework installation also.

All the versions of Visual Studio are backward compatible. We can open the project created in 2003 into VS2005 and it prompts you for the conversion using Visual Studio conversion wizard.

VS supports mainly three different languages, namely VB.NET, Visual C++, Visual C#. This IDE is supporting all object oriented features and drag and drop facility.

Solution and Project are the two main blocks of any .NET application which are developed using framework.

Solution is the container for the related projects. One solution can contain multiple projects. It allows us to specify settings and options that has to apply for multiple projects. Solution Explorer shows all information about your project. In ASP.NET, it refers to the website.

Project refers to the program created using Visual Studio. It contains the files that are interrelated. A project allows you to manage, build and debug the items that make an application. An application may have one or more projects.

When we build a project it usually results in the creation of an executable file or dynamic link library. These files that are created as a result of building the project are called project output.

✱ 2.2 ELEMENTS OF IDE

The IDE consists of the following components:

- **Windows forms designer:** It allows you to add controls to a form, arrange them and add code to perform some action. It provides you rapid development solution for an application.
- **Properties Window:** This is similar to the Visual Studio 6.0. It lists the properties associated with each and every control. This can be arranged alphabetically.
- **Solution Explorer:** This displays the solution name, project name, form name, module name and all other files associated with the project.
- **Tool Box:** It is provided with all types of controls. They can be easily dragged and dropped in the form. The toolbox has been divided into different categories of controls like Windows forms, data, Components, Containers, Crystal Reports, Menus and Toolbars.
- **Object Browser:** It shows the members of an object that are available.
- **Task list window:** It displays a list of errors along with the source of the error.
- **Server Explorer:** It is a very useful tool for showing database connectivity and displaying the tables, views and procedures associated with that connection..
- **Dynamic help:** It provides you with context sensitive help.
- **Class view:** It displays the classes, methods and properties.
- **Code and Text editor window:** It allows you to enter and edit code and text.

2.3 PROJECT TEMPLATES

The following templates are supported by VB.NET.

Windows Application: A project for creating an application with a Windows user interface.

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2.3 PROJECT TEMPLATES

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Windows Application: A project for creating an application with a Windows user interface.

Class Library: It is used for creating classes to use in other applications.

Console Application: It is used for creating command line application.

Windows Control Library: A project for creating controls to use in Windows applications.

Device Application: A project for creating a .NET Compact Framework 2.0 forms application for Pocket PC 2003.

Crystal Report: A project for creating a VB application with a Windows user interface and a sample Crystal Report.

2.4 WRITING A SIMPLE APPLICATION USING .NET

Click on the start button -> Visual Studio 2005. It opens the start page.

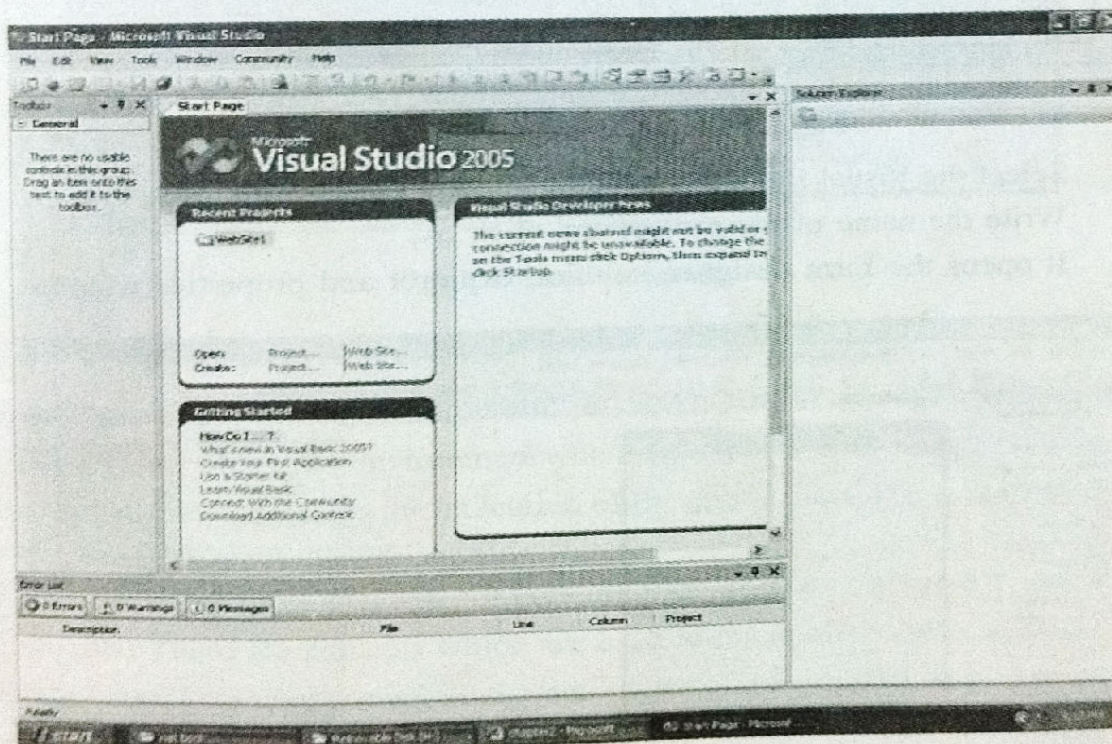


Fig. 2.1 Start Page.

Open the Visual Studio.Net and select create New project from the start page.

This opens the open project dialog box and shows the different types of project types.

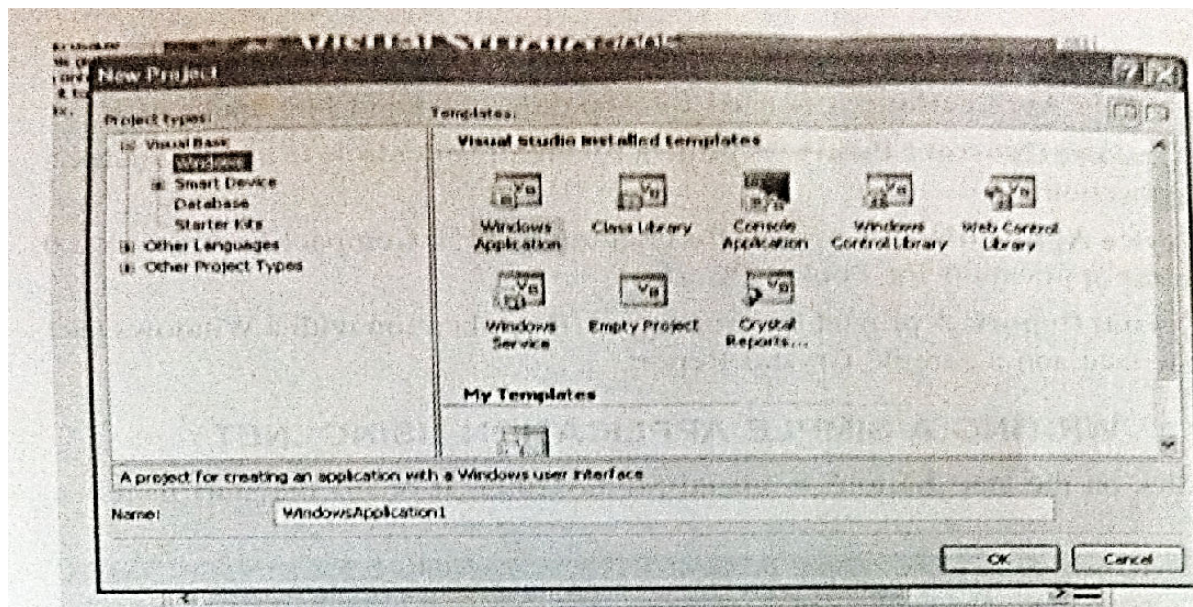


Fig. 2.2 Templates of Visual Basic.NET.

Select the Visual Basic project and Windows application template.

Write the name of the project and click ok.

It opens the form designer, solution explorer and properties window.

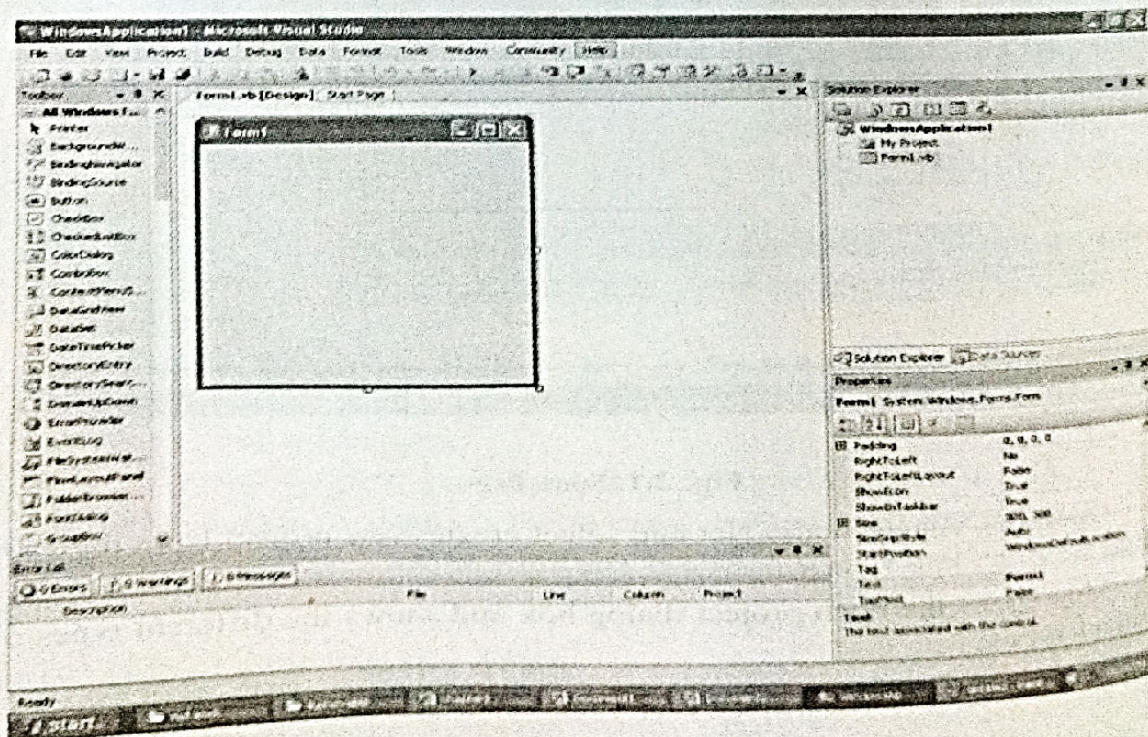


Fig. 2.3 Visual Studio IDE.

Design the form using different controls and write the code using code window.

Then save the project by selecting File -> save option.

To run the project press F5 or start debugging option.

It opens the form in the run mode.

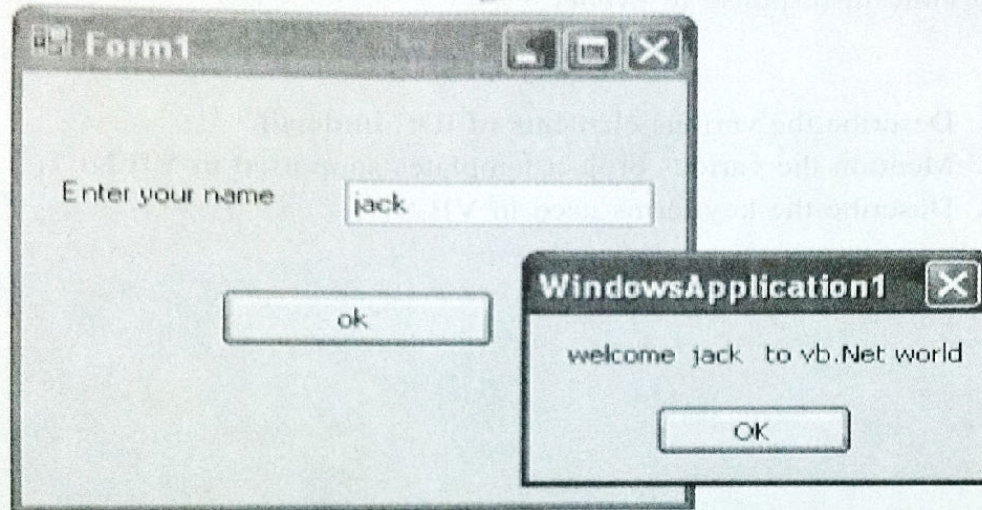


Fig. 2.4 Runtime mode.

2.5 VISUAL BASIC TERMS

The following key terms are important for developing a VB.NET application.

Design Time: This is the environment where the application is being developed.

Runtime: This represents the execution of the application. The user interacts with the application during runtime.

Forms: It is the basic element for creating user interface for an application.

Controls: These are controls which are used to manipulate the information.

Properties: These are known as characteristics of a control such as Caption, Color, Size etc.

Methods: The actions that can be performed by the object and on the object.

Events: These are actions that can be recognized by a form or control. These can be generated by the user, operating system or application interacts with the controls of a program.

POINTS TO REMEMBER

- Visual Studio.NET is an integrated development environment for all the languages such as Visual Basic, C++ and C#.
- This environment supports Windows and Web Forms for developing